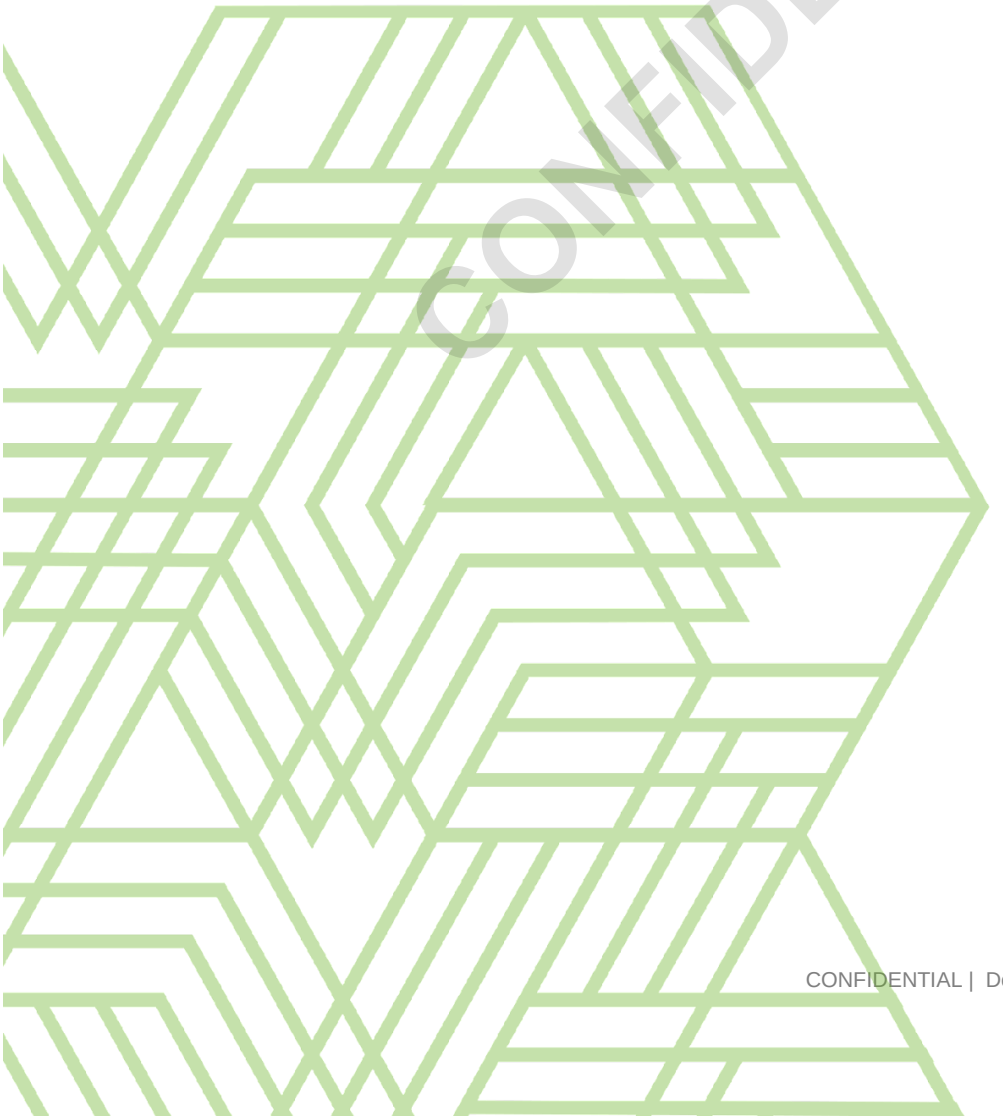
ATM2/ATM3/ATM33 Wireless SoC Family SDK

Quick Start Guide

SUMMARY: This document provides instructions on how to install the Atmosic Software Development Kit (SDK) and use it with the Atmosic Evaluation Kit (EVK) based on ATM2/ATM3 and ATM33 Wireless SoC Devices.

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1. Overview

This document provides instructions on how to install the Atmosic software development kit (SDK) and use it with an evaluation kit (EVK) based on the ATM2/ATM3/ATM33 wireless SoC Devices. For EVK power consumption measurements, please refer to the EVK Power Consumption Evaluation document for ATM2/ATM3 or ATM33. For more information about using the EVK, please refer to the EVK User Guides..

Refer to IAR Workbench User Guide, SEGGER Embedded Studio User Guide and Keil MDK Configuration Guide for installation and configuration if these tools are used with the SDK, available on the Atmosic Support website.

2. EVK Setup

2.1 ATM2/ATM3 EVB

1) Plug the interface board (**green**) into the EVB (**red**) as shown in Figure 1.
2) Attach the 2.4 GHz whip antenna to the RFIO port of the EVB as shown in Figure 2.
3) Plug in a USB cable into the USB1 port of the interface board as shown in Figure 3.
4) Plug the other end of the USB cable into a Windows computer.

Note: Newer versions of the EVB are **green** or **red**. The newer version of the interface board is **red**.

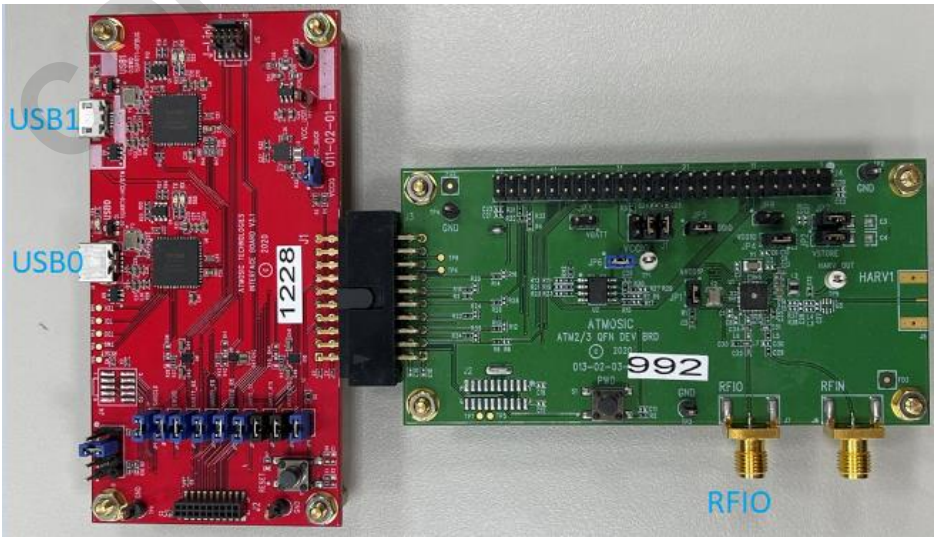


Figure 1 - Side-mount EVB To The Interface Board

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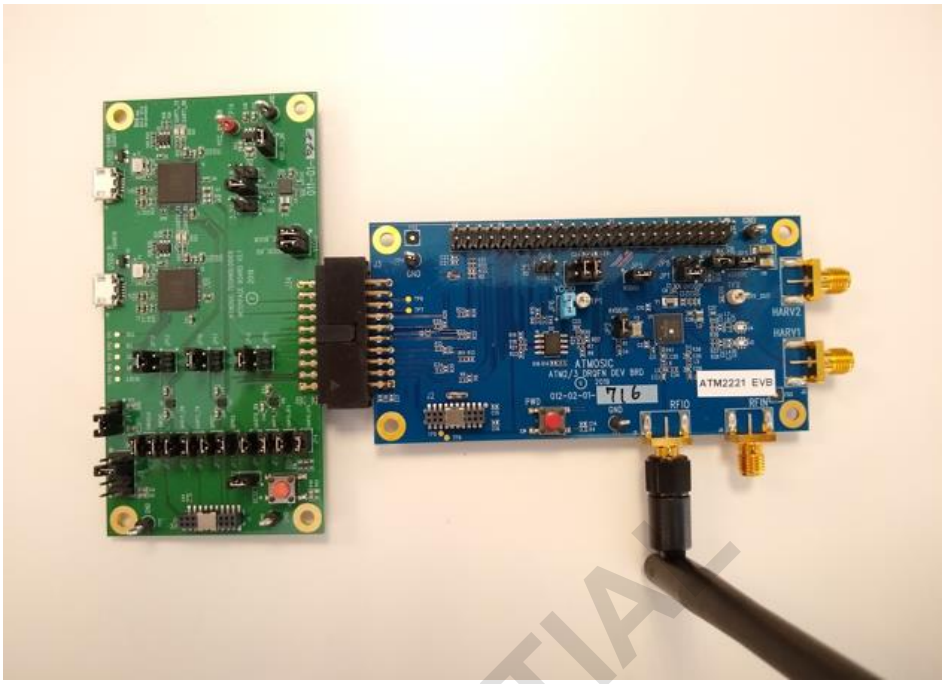


Figure 2 - EVB with 2.4 GHz Whip Antenna Attached

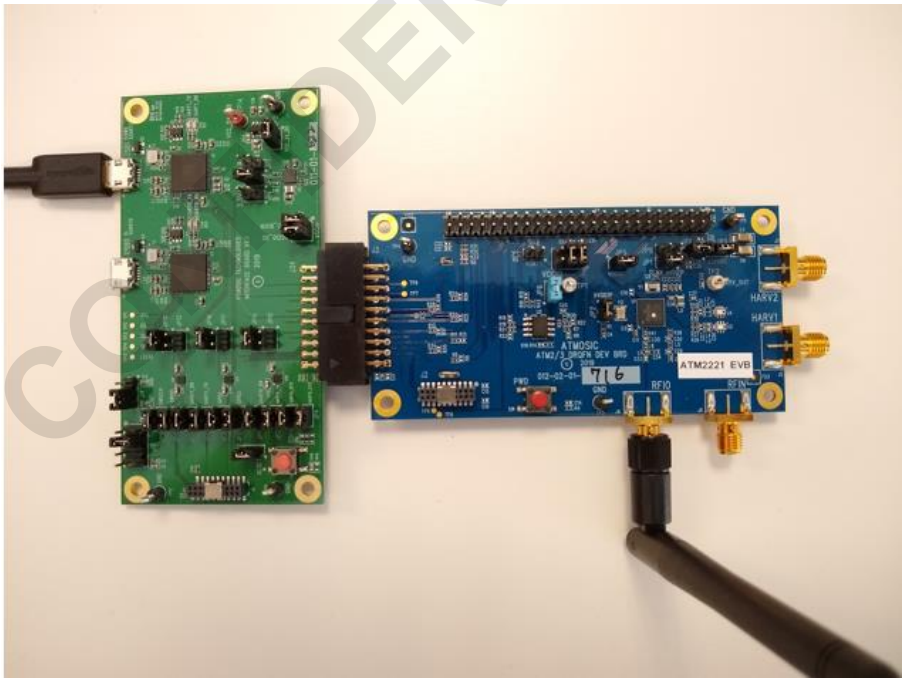


Figure 3 - Interface Board with USB Cable Attached

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2.2 ATM33 EVB

1) Plug the ATM3330e/ATM3330 EVB as shown in Figure 4 into the Windows computer using the Micro-B USB connector.
2) Attach the 2.4 GHz chip antenna to the SMA connector (labeled RFIO port on the board).

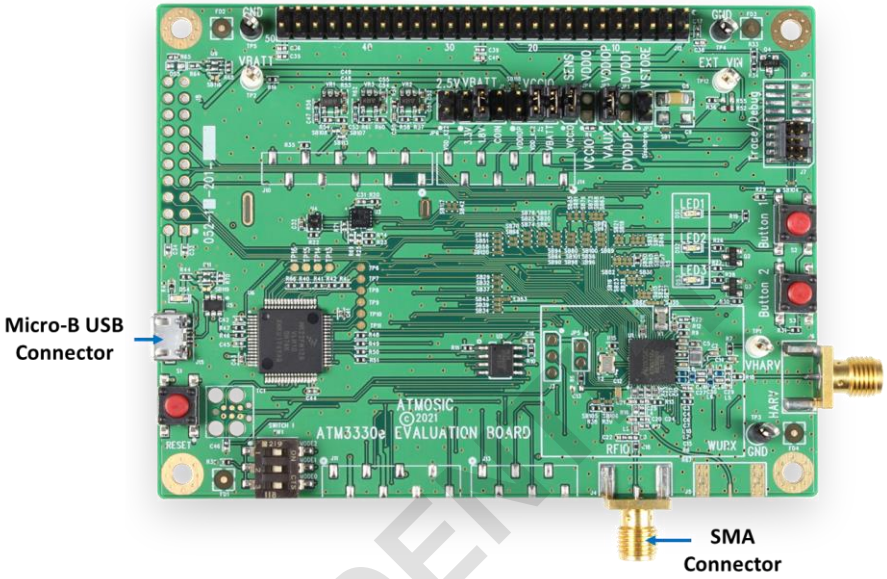


Figure 4 - ATM3330e Evaluation Board

3. SDK Installation

3.1 Using Windows Installer

The Windows 10 installer is available on the Atmosic Support website. The installation requires administrator privileges.

3.2 Install Atmosic SDK

The setup package of Windows Installer for Atmosic SDK is a compressed file named AtmosicSDK_Inst_[version_number].zip. Please extract the file and double click the “AtmosicSDK_Inst_[version_number].exe" to start the Atmosic SDK Installer. Figure 5 shows the release package source tree after the files are extracted.

AtmosicSDK_Inst_SDK_x.y.z	11/5/2020 2:39 PM	Application	426,061 KB
setup	11/5/2020 2:28 PM	Configuration settings	1 KB

Figure 5 - Extracted Release Package

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The Atmosic SDK will occupy 2.1 GB disk space and be located in the "C:\AtmosicSDK\" folder. See Figure 6. Please make sure the disk space is enough before installing the Atmosic SDK. Subsequently, click the "Install" button to continue.

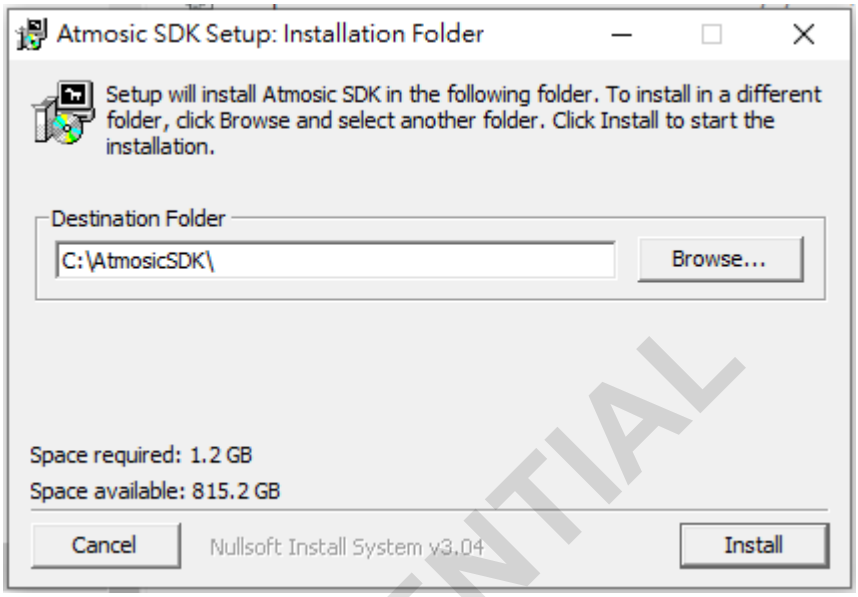


Figure 6 - SDK Setup Directory

Note: The installer cannot accept a path destination directory name with spaces. For example, "Atmosic SDK" is not acceptable while "AtmosicSDK" is acceptable. The Windows Installer will pop-up a message if the user selects an installation path name that includes spaces. See Figure 7.

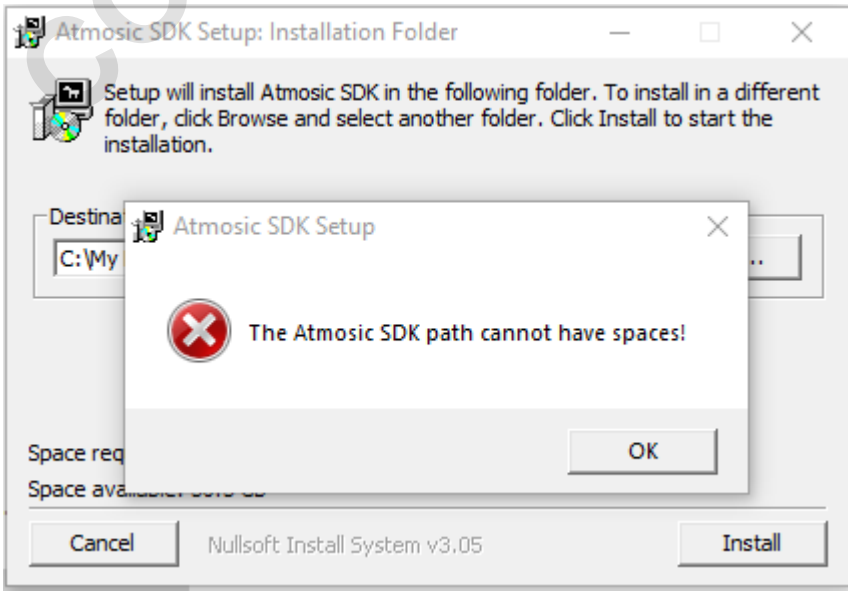


Figure 7 - SDK Installation Path Message

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The installation process takes several minutes to complete. The Windows Installer is installing MSYS2, GNU Toolchain and Atmosic SDK files to the target directory. See Figure 8.

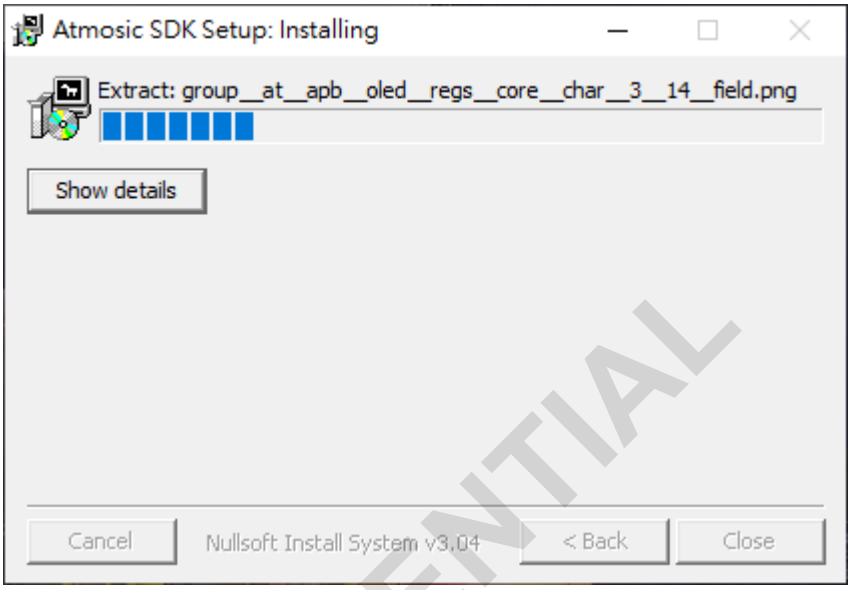


Figure 8 - Installing SDK

The Windows Installer will pop up a message to notify the installation completed, click “OK“ to close the Windows Installer. See Figure 9.

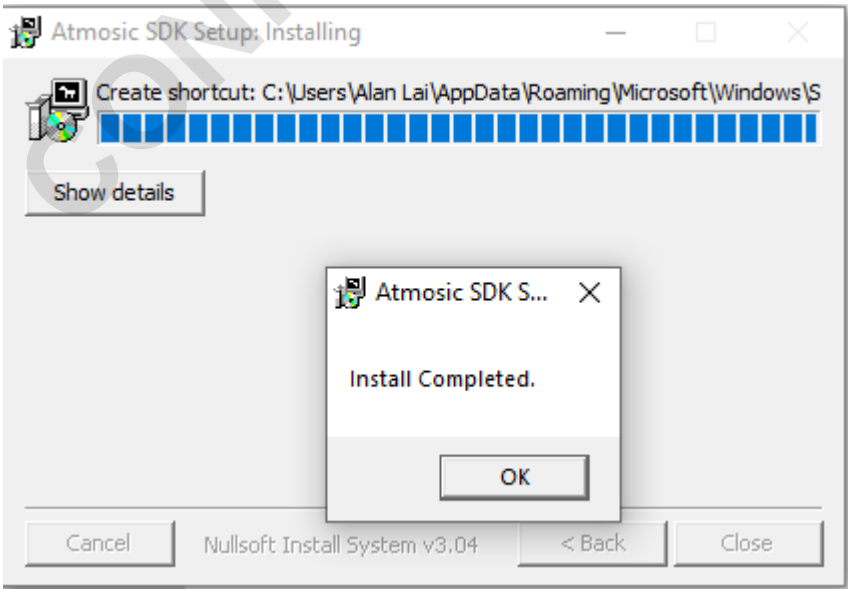


Figure 9 - Installation Completed

The Installer will create a folder named AtmosicSDK in “Start Menu” as shown in Figure 10 during the installation process.

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Figure 10 - SDK in Start Menu

3.3 Install RDI Interface

3.3.1 ATM2/ATM3 EVK

Please connect the Atmosic EVK (USB1) to the laptop and wait a while (~ 30 seconds).

Open the device manager (open the “Run” dialog box by pressing and holding the Windows key, then press the R key.

Enter “devmgmt.msc” then click the OK button to verify whether the Atmosic RDI USB interface exists or not.

If there are 2 COM ports listed in the device manager after plugging-in the Atmosic EVK to the laptop, please click the “Install RDI Interface" shortcut in “Start Menu”/Atmosic folder.

Note: The COM port numbers are assigned by the Windows system.

Note: If the device manager shows Atmosic RDI USB driver is already installed, move to the Connect To Debug Interface section.

Note: Atmosic RDI USB driver installation is required when the new interface board is connected to the computer at the first time.

Install Device Manager before installing Atmosic RDI USB driver.

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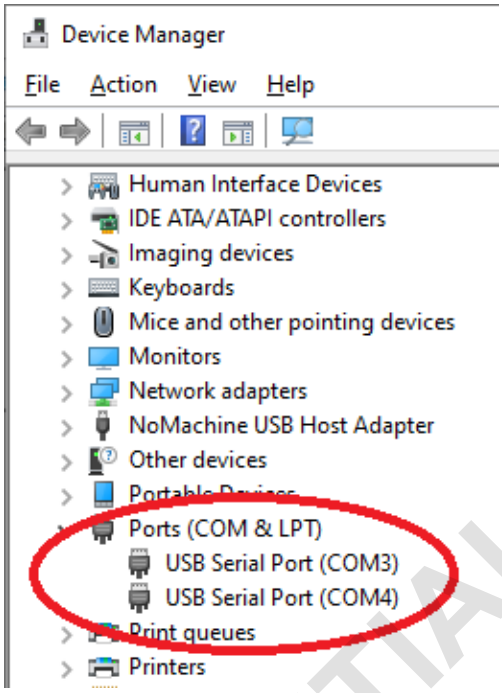


Figure 11 - USB COM Port

Please click the “Install” button if the system pop-up the following message.

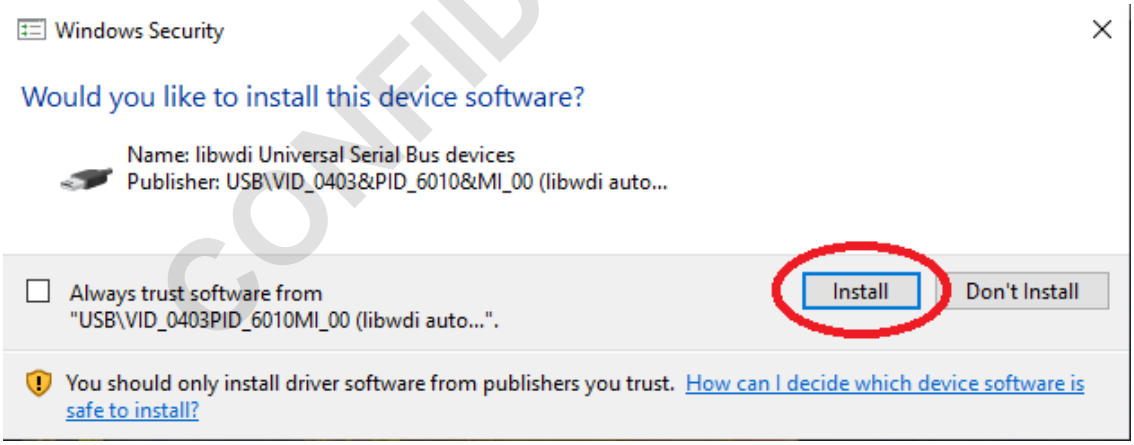


Figure 12 - Install RDI USB Driver

After installing Atmosic RDI USB driver, the Atmosic RDI USB1 interface will display in the Device Manager. Moreover, the COM4 is a debug console that could be used to display debug messages through PuTTY.

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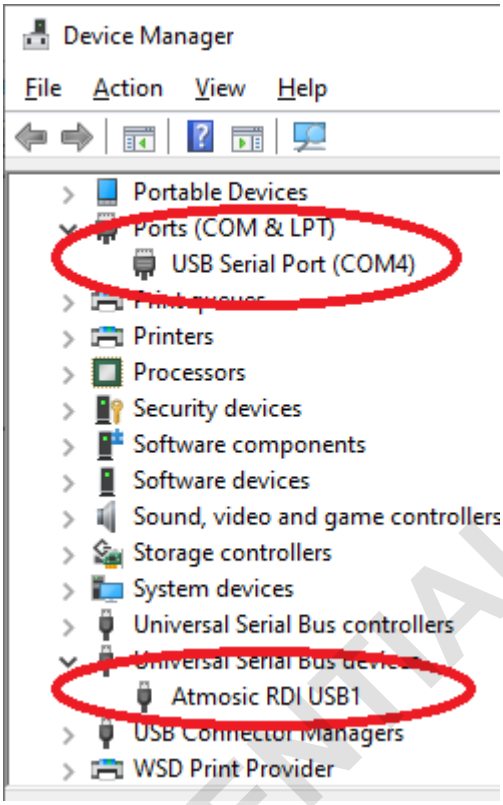


Figure 13 - RDI USB in Device Manager

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3.3.2 ATM33 EVK

Please connect the Atmosic EVK to the laptop and wait ~ 30 seconds.

Open the device manager (open the “Run” dialog box by pressing and holding the Windows key, then press the R key).

Enter “devmgmt.msc” then click the OK button to verify whether the J-Link OB interface exists or not.

If there are J-Link listed in the device manager of other devices, please click the “Install RDI Interface" shortcut in “Start Menu”/Atmosic folder.

Note: The COM port numbers are assigned by the Windows system.

Note: If the device manager shows “Install Completed”, move to the next section.

Open Device Manager before installing J-Link OB driver.

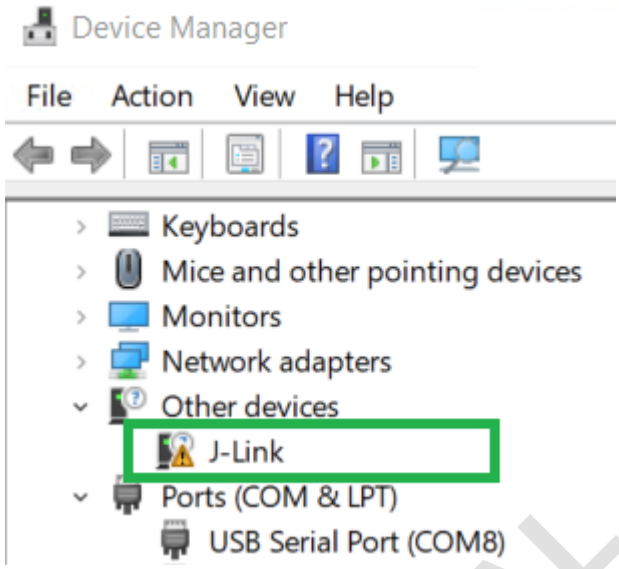


Figure 14 - USB COM Port

Please click the “Install” button if the system pop-up the following message.



Figure 15 - Install RDI USB Driver

After installing Atmosic RDI USB driver,verify the successful installation of the Windows
Device Manager and confirming that the "BULK interface" shows as such rather than "J-Link driver". (In Device Manager, expand category "Universal Serial Bus devices" and look for "BULK interface"). Moreover, the COM11 is a debug console that could be used to display debug messages through PuTTY.

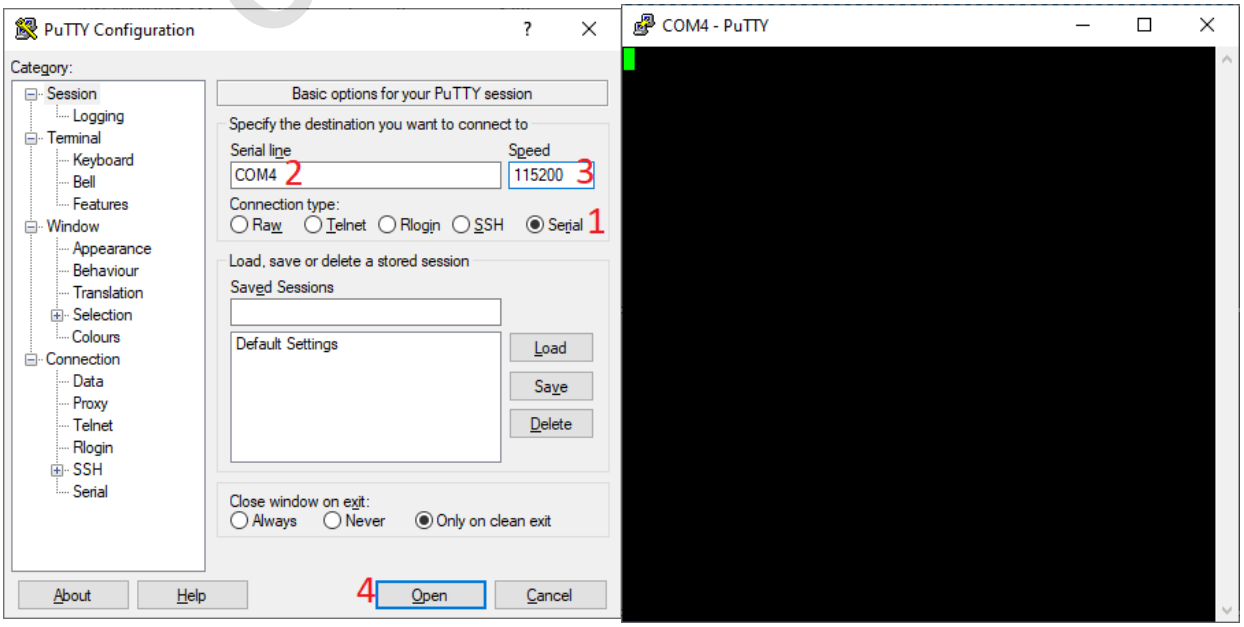
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Figure 16 - RDI USB in Device Manager

3.4 Connect to Debug Interface

The Atmosic EVK provides the log output through a debug interface (COM port). The developer could use utility to capture (i.e. PuTTY, Tera Term ...etc.) log. This section describes how to capture logs through PuTTY. When open the PuTTY tool, click the "Serial" radio button then input the COM port information in "Serial line" item (i.e. COM11), and input 115200 in "Speed" item. Subsequently, click the "Open" button to open the COM port and wait for debug message input. See Figure 17.



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Figure 17 - Debug Interface Using PuTTY

3.5 Launch MSYS2 shell

Open MSYS2 shell from Windows menu, “Start Menu” → ”AtmosicSDK” → ”SDK_[version_number]”. Figure 18 shows the Start Menu.



Figure 18 - Start Menu

The MSYS2 shell will open and redirect the working path to the Atmosic SDK path as shown in Figure 19.

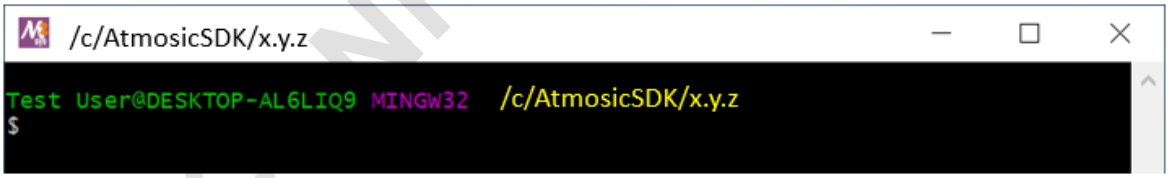


Figure 19 - Atmosic SDK Path

3.6 Atmosic SDK Platform Working Directory

SDK 5.1.0 supports ATM2, ATM3 and ATM33 devices. The difference between ATM2 and ATM3 device versions is captured by the second digit in the -xxx extension of the product code for the device.

In order to build the correct application software for any ATM2, ATM3 or ATM33 device it is necessary to identify the device version and use the appropriate directory in the Atmosic Software Development Kit (SDK) for the software build.

ATM2 and ATM3 evaluation boards have a sticker that identifies the part number of the IC on the board. For example, the Atmosic EVB is labeled with ATM3221-010 as

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shown in Figure 20. It is assembled with an ATM32xx-x1x chip and to compile the project, the path is platform/atm3/ATM32xx-x1x. Refer to the Atmosic Chip Revisions Application Note for information regarding chip revisions.

The ATM33 evaluation board has printed the part number of the IC on the board, please refer to Figure 4.

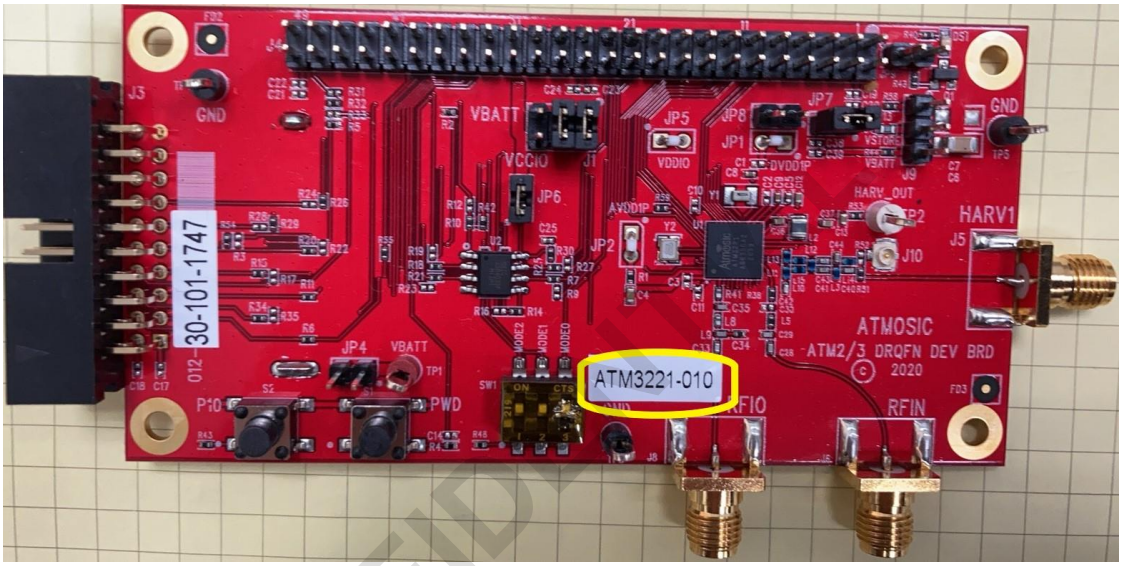


Figure 20 - Atmosic Product Code on the EVB

Following is a list of all Atmosic SDK platform working directories, please use the appropriate directory in the Atmosic SDK for the software build.

- platform/atm2/ATM22xx-x0x
- platform/atm2/ATM22xx-x1x
- platform/atm3/ATM32xx-x0x
- platform/atm3/ATM32xx-x1x
- platform/atm33/ATM33xx-3

3.7 Simple Beacon Configurator

Atmosic EVK comes with a pre-programmed Beacon application in Flash. Simple beacon configurator web tool allows the user to change/tune a few parameters of this beacon. This tool allows beacon configuration as Eddystone or iBeacon or Beacon with Manufacturer specific data. See Figure 21.

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Note: The Simple Beacon Configurator is only meant for use with ATM2201 and ATM2221 EVK's. For ATM3201 and ATM3221 EVK's, please skip to Building ATM3 Beacon Application Using Atmosic SDK section.

To use the simple beacon configurator, run following commands in MSYS2 shell

```
$ cd platform/atm2/ATM2xx-x0x/examples/BLE_adv/
$ make openocd
```

Now open following page from favorite browser (replace <version_number> with the current SDK version):

```
C:\AtmosicSDK\<version_number>\tools\configurator\index.html
```

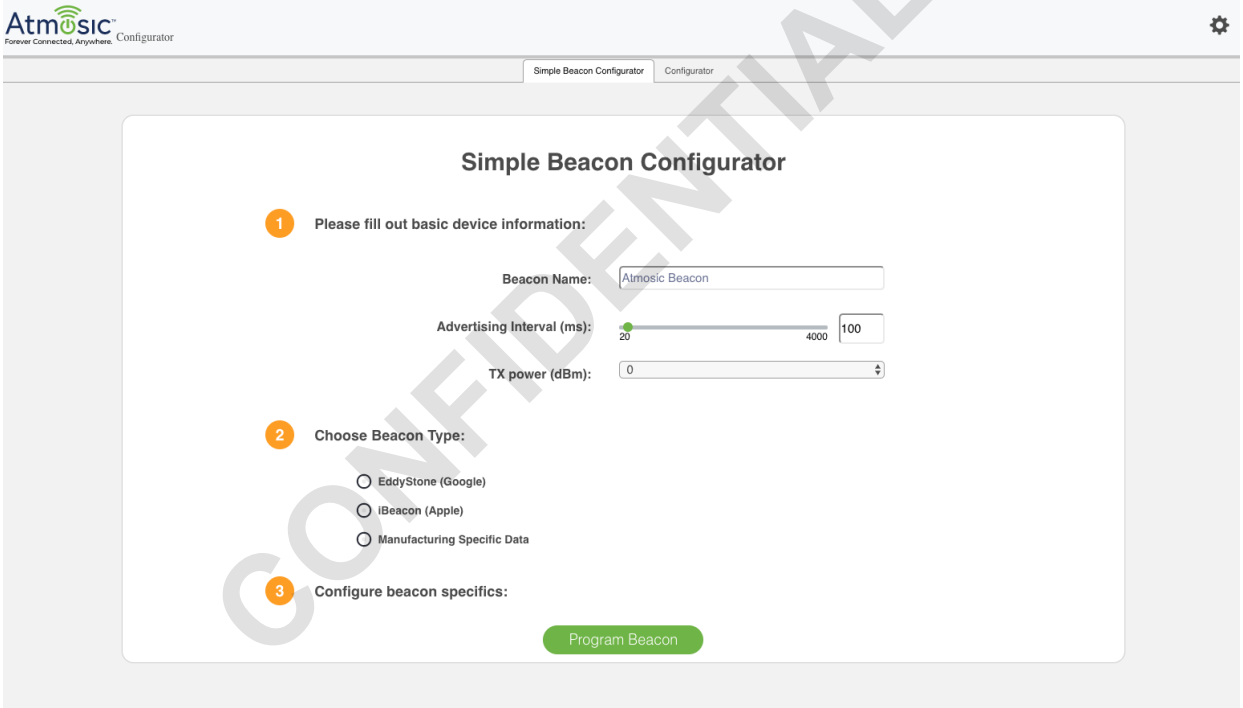


Figure 21 - Simple Beacon Configurator

Fill out all sections as needed (preset defaults are used for unchanged fields).
Once configuration is done, click on the “Program Beacon” button to program EVK with updated beacon parameters.

Note: Simple Beacon configurator does not restore previous settings and starts with preset defaults.

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4. Building Beacon Application Using Atmosic SDK

4.1 ATM2 Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project through the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval.

```
$ cd platform/atm23/ATM23xx-x1x/examples/BLE_adv/
$ make
```

If everything is correct, the “make” should execute without error as shown in Figure 22.

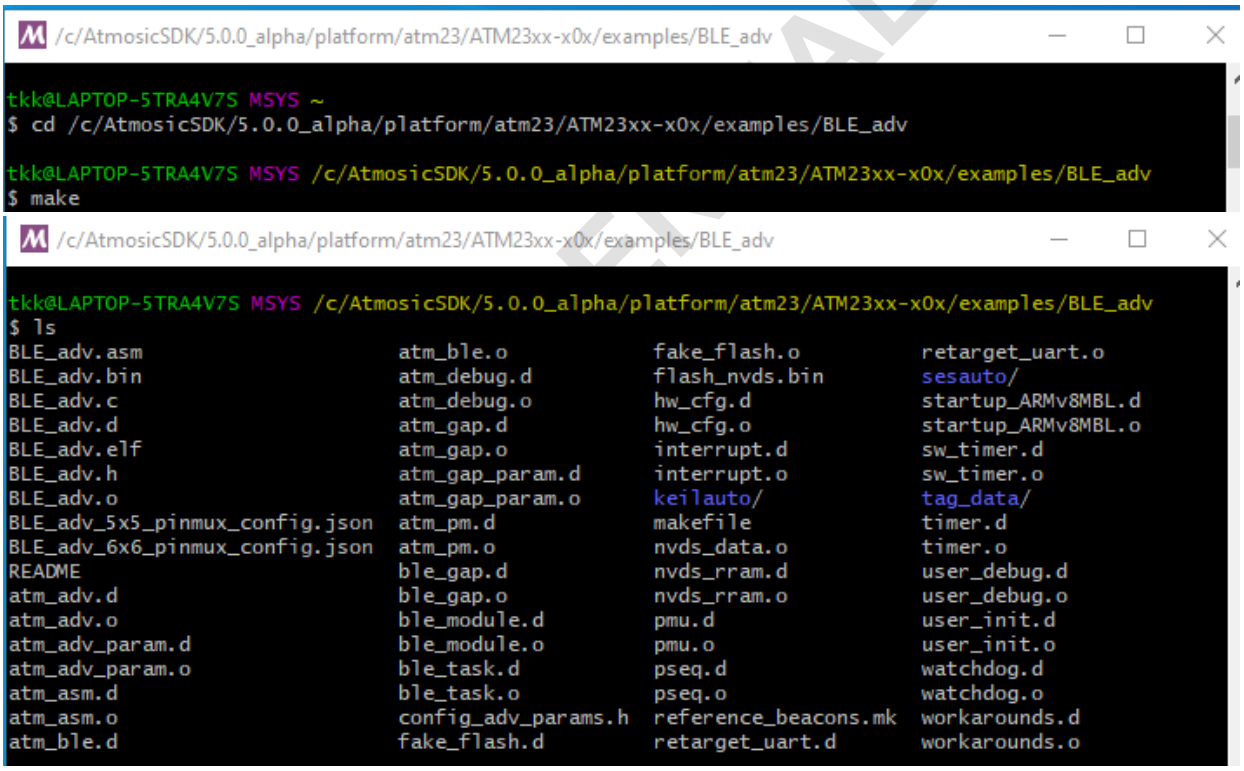


Figure 22 - Make Executed without Error

At MSYS2 shell, type “make run_all” to build and load firmware into Atmosic EVK. Figure 23 shows the results on shell and debug terminals after the “make run_all” command is successfully executed.

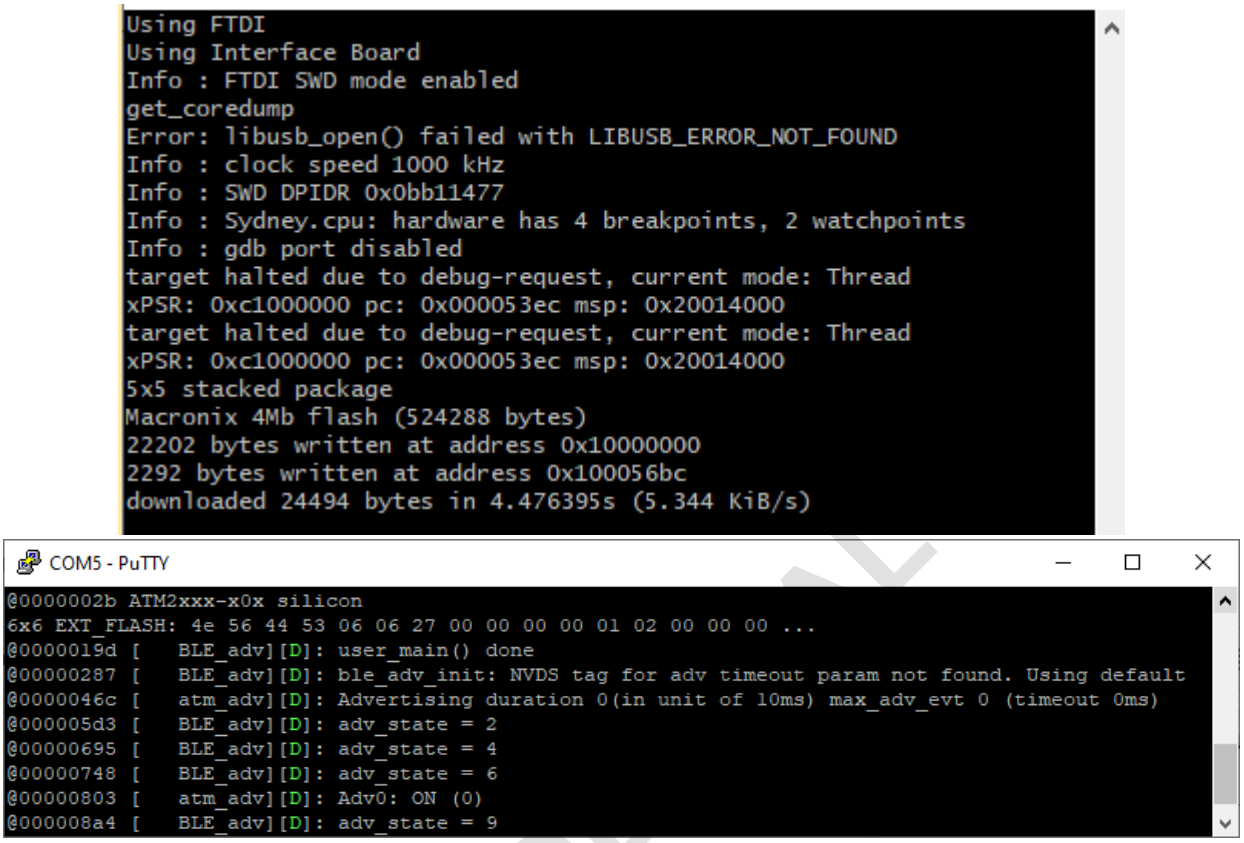


Figure 23 - Execution result of "make run_all"

Detailed documentation for Beacon and other examples in SDK is available in platform/atm2/ATM2xx-x0x/examples/<example_name>/README file.

4.2 ATM3 Device

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project through the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm3/ATM32xx-x0x/examples/BLE_harv_adv/
$ make DEBUG=  run_all
```

4.3 ATM33 Device

At MSYS2 shell, go to the BLE_adv example folder to compile the project using the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 100 ms Interval.

```
$ cd platform/atm33/ATM33xx-3/examples/BLE_adv/
$ make run_all
```

5. Building ATM3330e Energy Harvesting Beacon Application Using Atmosic SDK

At MSYS2 shell, go to the BLE_harv_adv example folder to compile the project through the following command.

Beacon configuration: EddyStone, Non Connectable, Scannable, 1s Interval.

```
$ cd platform/atm33/ATM33xx-3x/examples/BLE_harv_adv/
$ make DEBUG=  run_all
```

6. Uninstall Atmosic SDK

Follow the steps to uninstall Atmosic SDK. Also see Figure 24.

1) Click the “Start Menu”/”Settings”.
2) In the “Settings” page, click the “Apps” item.
3) Find the “Atmosic SDK (<version_number>)” item in the “Apps & features” page
4) Click the “Uninstall” button to start the Atmosic SDK uninstallation process.
5) The Atmosic RDI Interface will be removed during the uninstallation process. Please pull-out the Atmosic EVK before reinstalling the Atmosic SDK again.

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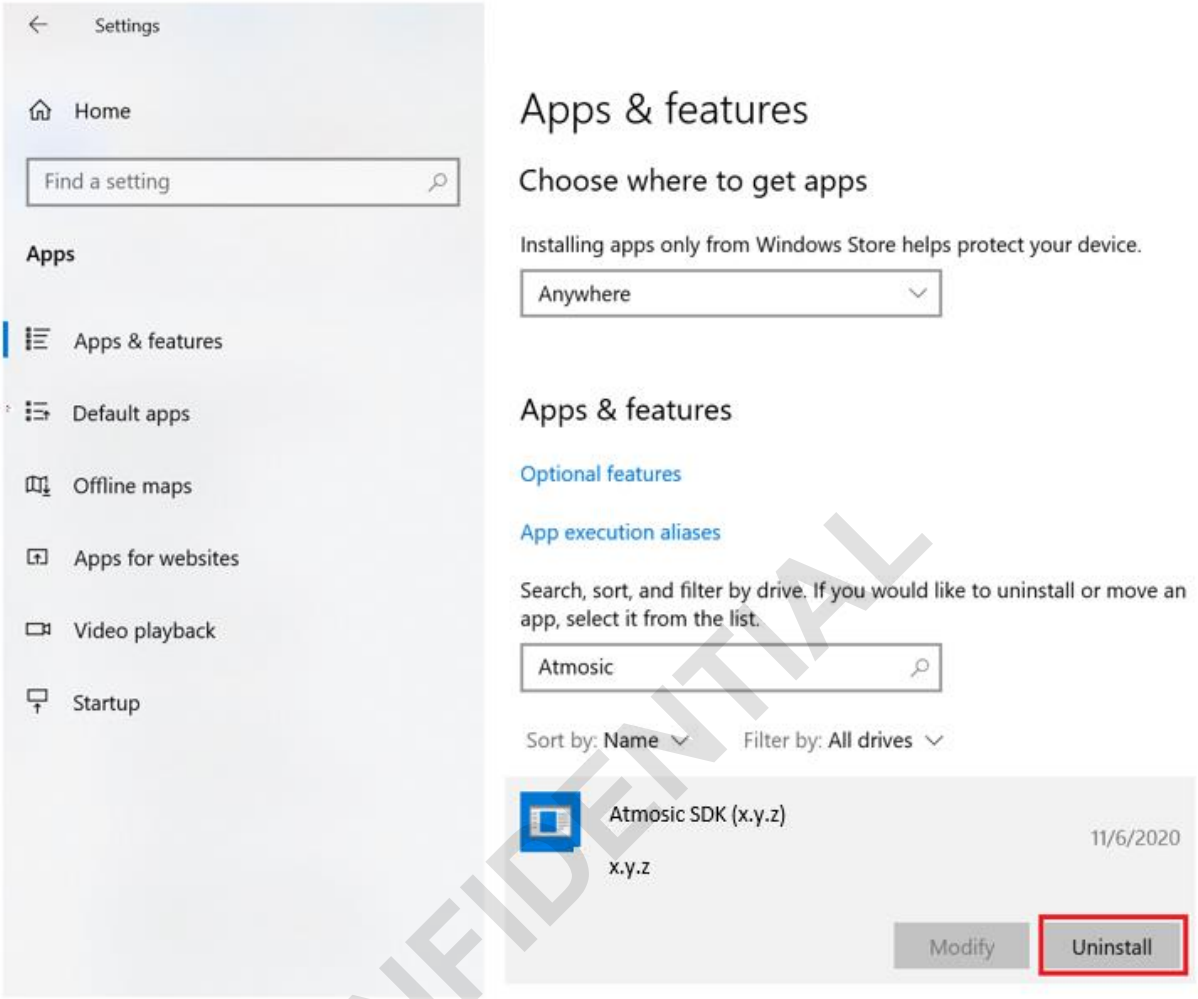


Figure 24 - Uninstall SDK

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Revision History

Date	Version	Description
May 13, 2022	0.60	Initial version created for SDK 5.1.0 release.

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